

NUST

NATIONAL UNIVERSITY OF SCIENCES & TECHNOLOGY

Name: Muhammad Abdullah

Department/Batch: IGIS-2024

CMS Id: 501223

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Unsupervised Classification Lab Report (ERDAS Imagine)

Objective

Perform unsupervised classification on satellite imagery using K-Means and ISODATA in ERDAS Imagine, then reclassify into five land-cover categories.

Task 1: Unsupervised Classification

1.1 K-Means Classification in ERDAS Imagine

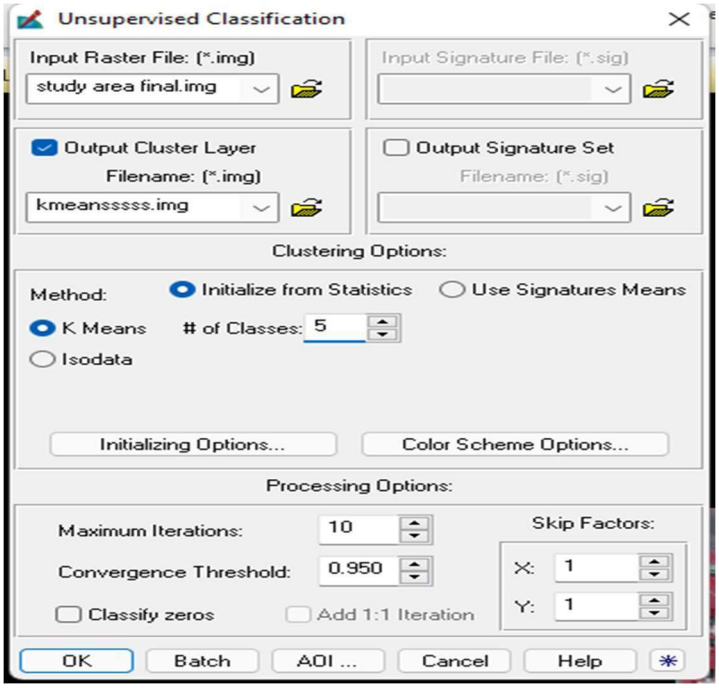


Figure 1



Figure 2

1.2 How K-Means Works

K-Means partitions pixels into K clusters by:

- Randomly placing K cluster centers in spectral space
- Assigning each pixel to nearest center
- Recalculating center positions
- Repeating until cluster centers don't move (convergence)
- Final output: each pixel labeled with cluster number

1.3 ISODATA Classification in ERDAS Imagine

Steps in ERDAS Imagine:

1. Open satellite image → Raster → Unsupervised Classification → ISODATA
2. Set initial number of clusters
3. Set merge and split parameters
4. Set convergence criteria
5. Run classification

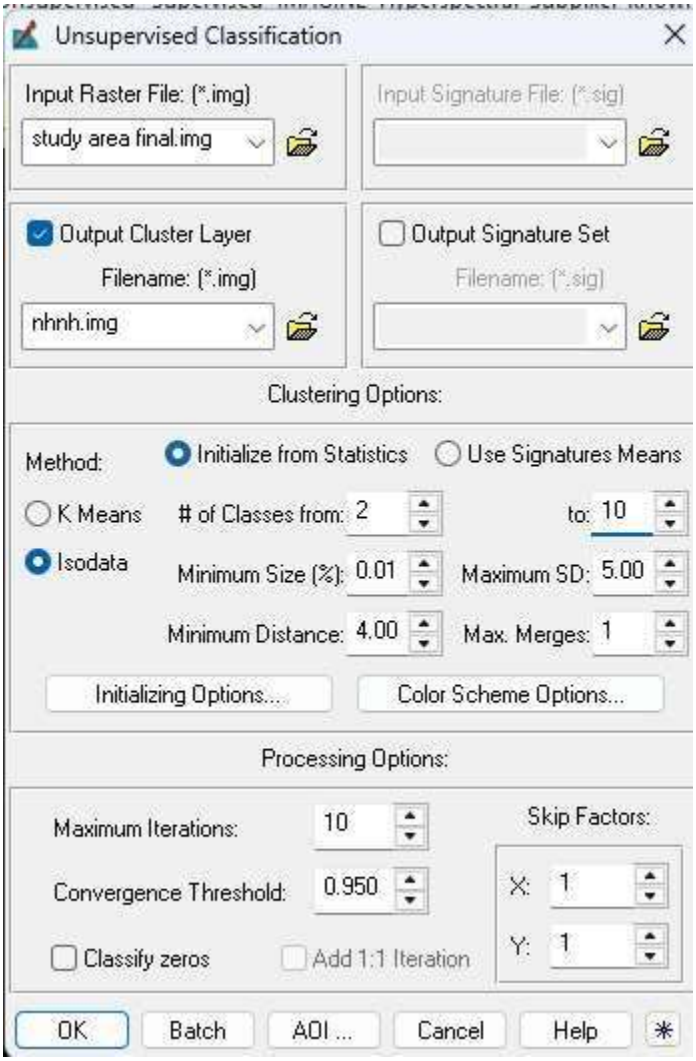


Figure 3

1.4 How ISODATA Works

ISODATA automatically adjusts clusters by:

- Starting with initial clusters (like K-Means)
- Calculating cluster statistics (mean distance, standard deviation)
- **Merging:** Combines clusters closer than threshold distance
- **Splitting:** Divides clusters with high standard deviation
- Repeating until stable (cluster count may change)
- Final output: optimized number of clusters



Figure 4

1.5 Comparison: K-Means (5 classes) vs ISODATA (10 classes)

How K-Means works: I set K=5, algorithm forces pixels into exactly 5 groups. Simple and predictable.

How ISODATA works: Starts with estimate, automatically merges/splits clusters. Ended with 10 classes because algorithm found more natural groupings in data.

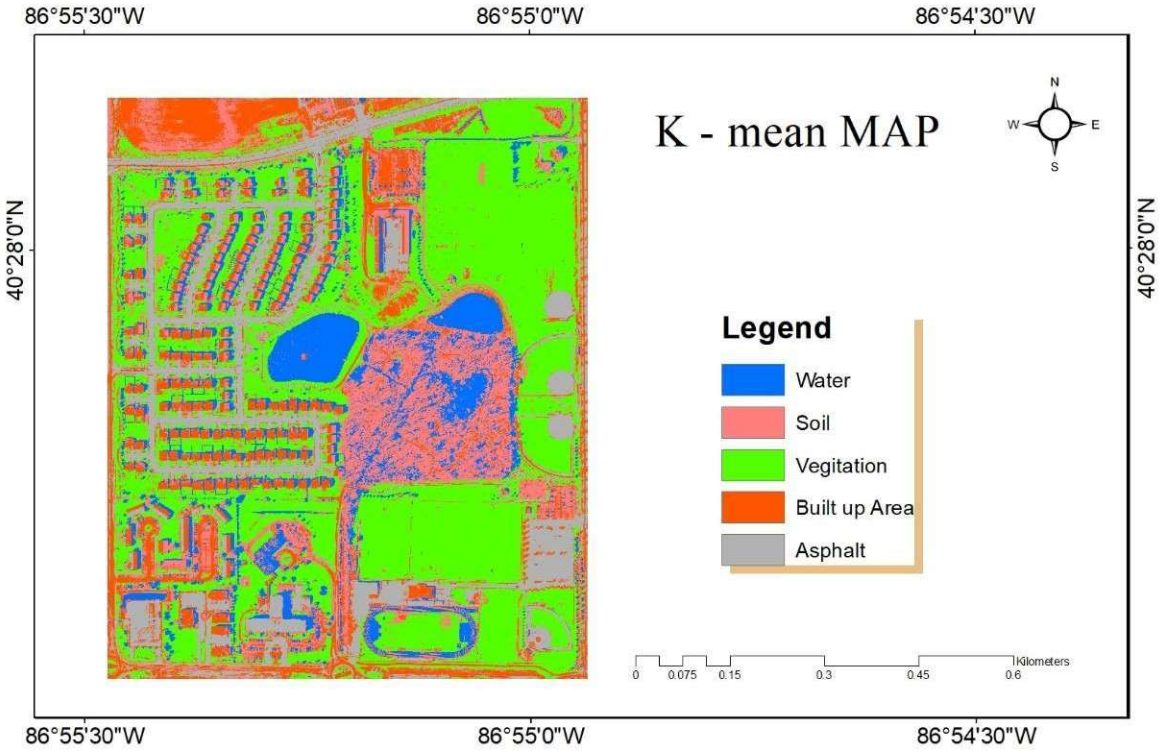
Key Difference:

- K-Means: Fixed at 5 classes (you control)
- ISODATA: Adaptive to 10 classes (algorithm controls)

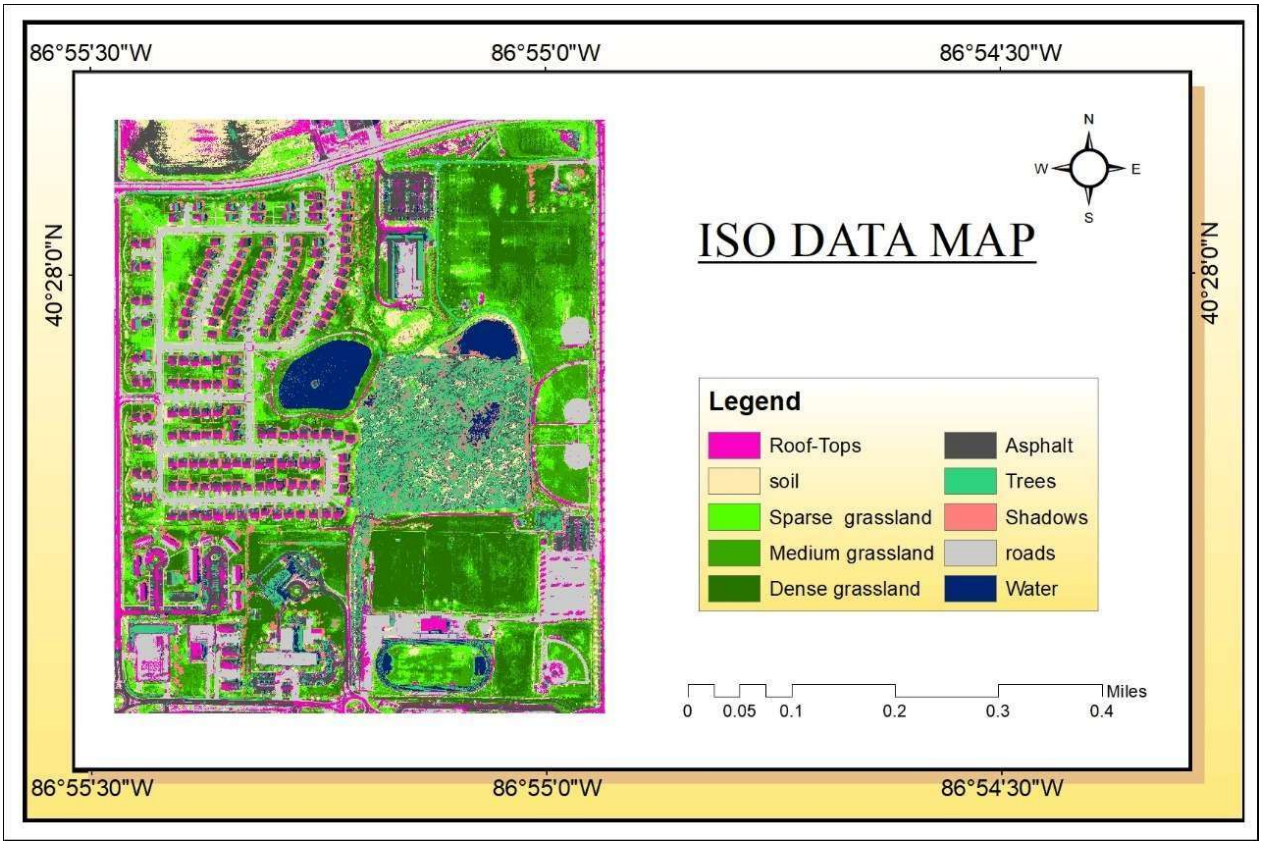
ISODATA gave more detail by discovering additional classes that K-Means missed because it was locked at 5.

Task 2: Reclassification

K-Means:



ISO-DATA:


86°55'30"W
86°55'0"W
86°54'30"W
40°28'0"N
N
W
E
S
ISO DATA MAP
Legend
Roof-Tops
soil
Sparse grassland
Medium grassland
Dense grassland
Asphalt
Trees
Shadows
roads
Water
0
0.05
0.1
0.2
0.3
0.4
Miles